

Problems Week 1

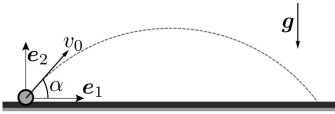
Problem 1: Projectile Motion

We consider the same system as in Clicker 2. A particle is launched at time $t = 0$ with an initial velocity v_0 under an angle α with the horizontal axis, as shown. During its flight, the particle is subject to gravity $\mathbf{a} = -g\mathbf{e}_2$. (We neglect the influence of air.)

Given: v_0, α, g

$$\underline{v}(t=0) = \begin{pmatrix} v_0 \cos \alpha \\ v_0 \sin \alpha \end{pmatrix}$$

$$\underline{r}(t=0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$



$$\underline{F} = M \cdot \underline{a}$$

← kinetics

- (a) Compute the trajectory $\underline{r}(t)$ of the particle. When and where does it hit the ground?
- (b) How do the result and the computations change when considering air? What changes when considering that the gravitational acceleration depends on altitude? You do not need to compute the result.

- in vacuum.
- immersed in a medium causing a viscous drag equivalent to $\mathbf{a}_{\text{drag}}(\mathbf{v}) = -k\mathbf{v}$.
- in vacuum and considering the change of gravity with altitude given by $\mathbf{a}(\mathbf{r}) = -g \left(\frac{R}{R+r_2} \right)^2 \mathbf{e}_2$ whereas R is the radius of the earth.

$$a) \quad \underline{F} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2 = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ -Mg \end{pmatrix}$$

← split up in components!

$$\underline{a}(t) = \ddot{\underline{x}}(t) = \begin{pmatrix} 0 \\ -g \end{pmatrix}$$

$$\hookrightarrow \underline{v}(t) = \int \underline{a}(t) dt = \begin{pmatrix} c_1 \\ -gt + c_2 \end{pmatrix} = \begin{pmatrix} v_0 \cos \alpha \\ v_0 \sin \alpha - gt \end{pmatrix}$$

$$\text{IC: } \begin{aligned} \dot{x}_1(0) &= v_0 \cos \alpha \\ \dot{x}_2(0) &= v_0 \sin \alpha \end{aligned}$$

$$\underline{r}(t) = \int \underline{v}(t) dt = \begin{pmatrix} v_0 \cos \alpha \cdot t + k_1 \\ v_0 \sin \alpha \cdot t - \frac{1}{2}gt^2 + k_2 \end{pmatrix} \stackrel{\text{at } t=0}{=} \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{aligned} k_1 &= 0 \\ k_2 &= 0 \end{aligned}$$

to hit ground

$$v_0 \sin \alpha \cdot t - \frac{1}{2}gt^2 \stackrel{!}{=} 0 \quad | : t$$

$$v_0 \sin \alpha = \frac{1}{2}gt \Rightarrow t = \frac{2v_0 \sin \alpha}{g} \Rightarrow \text{plug in } t \text{ into } \underline{r}(t)$$

↳

b) 2. $\underline{x}_1$: $\ddot{x}_1(t) = -k \dot{x}_1(t)$

$$\frac{dx_1}{dt} = -k x_1$$

$$\int_{v_0 \cos \alpha}^{x_1} \frac{1}{x_1} dx_1 = \int_0^t -k dt$$

$$\hookrightarrow x_1 = v_0 \cos \alpha e^{-kt}$$

$$x_1(t) = x_1(0) + \int_0^t \dot{x}_1(t) dt$$

$$x_1(t) = \frac{v_0 \cos \alpha}{k} (1 - e^{-kt})$$

$\underline{x}_2$: $\ddot{x}_2(t) = -k \dot{x}_2 - g$ (1)

$$\frac{du}{dt} = -k \dot{x}_2 - g \quad (\ddot{x}_2 = u)$$

K: $u(t=0) = -k v_0 \sin \alpha - g$

$$\int_{-k v_0 \sin \alpha - g}^{u(t)} \frac{1}{u} du = \int_0^t -k dt$$

$$u(t) = -(k v_0 \sin \alpha + g) e^{-kt}$$

$\hookrightarrow$ plug into (1)

$$\ddot{x}_2(t) = \left(v_0 \sin \alpha + \frac{g}{k} \right) e^{-kt} - \frac{g}{k}$$

now integrate: $x_2(t) = x_2(0) + \int_0^t \ddot{x}_2(\tau) d\tau$

$$= \left[\frac{v_0}{k} \sin(\alpha) + \frac{g}{k^2} \right] (1 - e^{-kt}) - \frac{gt}{k}$$

3. (don't worry)

$\underline{x}_1$: $\ddot{x}_1(t) = 0$, $\dot{x}_1(t) = v_0 \cos \alpha \xrightarrow{\int dt} x_1(t) = v_0 \cos \alpha \cdot t$

$\underline{x}_2$: $\ddot{x}_2(t) = -g \left(\frac{R}{R + x_2(t)} \right)^2$, where $v_2 = \dot{x}_2 \rightarrow \dot{v}_2 = \ddot{x}_2$

Approximate derivatives with finite differences = Euler method

$$\frac{x_{k+1} - x_k}{\Delta t} = v_k \implies x_{k+1} = x_k + v_k \Delta t$$

$$\frac{v_{k+1} - v_k}{\Delta t} = -g \left(\frac{R}{R + x_k} \right)^2 \implies v_{k+1} = v_k - g \left(\frac{R}{R + x_k} \right)^2 \cdot \Delta t$$

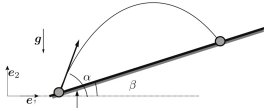
$\hookrightarrow$ solve iteratively (in program) with $x_0 = 0$, $v_0 = v_0 \sin \alpha$

... bla bla bla

Problem 2 : Projectile Motion #2

How does the previous problem change, if it is launched on an inclined ground of inclination angle β as shown below? (No need to solve the problem, simply explain what changes in the derivation.)

Given: v_0 , α , β , g

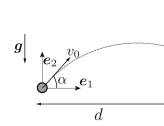


$$\hookrightarrow x_2 = x_1 \tan \beta$$

Problem 3 : Projectile Motion #3

How does the problem change, if the particle is launched against a wall at a distance d . When and where does it hit the wall? (No need to solve the problem, simply explain what changes in the derivation.)

Given: v_0 , α , β , d , g



$$\hookrightarrow x_1 = d$$

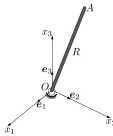
same procedure as in Problem 1)

Problem 4 : Guided motion

A rigid arm of length R rotates about a spherical hinge joint at point O , as shown below. The x_1 - and x_2 -coordinates describing the spatial motion of point A (the arm's tip) are recorded as, respectively,

$$x_1(t) = R \cos(\omega t), \quad x_2(t) = R \sin(\omega t) \cos(\omega t), \quad (1)$$

where ω is a known constant. Assume that the hinge is attached to a planar ground, so the motion is restricted to $x_3 \geq 0$.



(a) What is the position $\mathbf{r}(t)$ of point A at a given time t ?

(b) What is the velocity vector $\mathbf{v}(t)$ of point A at a given time t ?

Hint: Pythagoras

$$a) \quad x_3^2 = R^2 - x_1^2 - x_2^2$$

$$x_3^2(t) = R^2 - R^2 \cos^2(\omega t) - R^2 \sin^2(\omega t) \cos^2(\omega t) \\ R^2 [1 - \cos^2(\omega t) - \sin^2(\omega t) \cos^2(\omega t)]$$

(use $\sin^2 + \cos^2 = 1$)

$$x_3^2(t) = R^2 \sin^2(\omega t) (1 - \cos^2(\omega t)) = R^2 \sin^4(\omega t)$$

$$\hookrightarrow x_3 = R \sin^2(\omega t)$$

$$\hookrightarrow \mathbf{r}(t) = \begin{pmatrix} R \cos(\omega t) \\ R \sin(\omega t) \cos(\omega t) \\ R \sin^2(\omega t) \end{pmatrix}$$

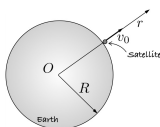
$$b) \quad \mathbf{v}(t) = \dot{\mathbf{r}}(t)$$

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Problem 5: Orbital dynamics.

A satellite is launched from the surface of the Earth with an initial radial velocity v_0 and it is subject to the gravitational radial acceleration $a_r = -g \frac{R^2}{r^2}$, where R is the radius of the Earth and r the distance from the center of the Earth. Neglect the Earth's rotation.

Given: R, g, v_0



(a) What is the velocity $v(r)$ of the satellite (for a given v_0)?

(b) What is the minimum initial velocity v_0 so that the satellite can leave the Earth's gravitational influence (i.e., travel to infinite distances)?

Hints: 1) $a = \frac{dv}{dt} = \frac{dv}{dr} \frac{dr}{dt} = \frac{dv}{dr} v$

2) estimate minimal velocity to escape Earth's gravitational force

$$v(r \rightarrow \infty) = \lim_{r \rightarrow \infty}$$

a) $a(r) = \frac{dv}{dt} = -g \frac{R^2}{r^2}$

using 1st hint: $\frac{dv}{dr} \frac{dr}{dt} = \frac{dv}{dr} v = -g \frac{R^2}{r^2}$
ODE

$$\int v dv = \int g \frac{R^2}{r^2} dr \Rightarrow \underline{\underline{v(r) = \sqrt{v_0^2 + 2gR \frac{R-r}{r}}}}$$

b) using 2nd hint: $v(r \rightarrow \infty) = \lim_{r \rightarrow \infty} \sqrt{v_0^2 + 2gR \frac{R-r}{r}} = \sqrt{v_0^2 + 2gR}$

Since we want $v(r \rightarrow \infty) = 0$ we get the following condition:

$$v_0^2 - 2gR > 0 \Rightarrow \underline{\underline{v_0 > \sqrt{2gR}}}$$

some of you might recognize from relativity theory in high school 😊